

Micro3: Production & Competitive Equilibrium

Session 1: Production, Profit Maximization, and Cost Minimization

1. Production plans, production sets, and production functions

- A **production plan** is a vector $y = (y_1, \dots, y_n) \in \mathbb{R}^n$ of outputs and inputs (outputs positive; inputs negative in the sign convention). Example (single output, two inputs): producing 1 unit of output using 2 units of apples and 3 units of water can be written as $y = (1, -2, -3)$.
- A **production set** $Y \subseteq \mathbb{R}^n$ collects all feasible production plans.
- **Single-product case:** write a plan as $(q, -z_1, \dots, -z_m)$ with $q \geq 0$ and $z \in \mathbb{R}_+^m$.
- Often assume a **production function** $f : \mathbb{R}_+^m \rightarrow \mathbb{R}_+$ such that output is bounded by $q \leq f(z)$.

2. Returns to scale and Euler equation

Let $f : \mathbb{R}_+^m \rightarrow \mathbb{R}_+$.

- **Nonincreasing returns to scale (NIRS):** for any z and any $\alpha \in [0, 1]$,

$$f(\alpha z) \geq \alpha f(z).$$

- **Nondecreasing returns to scale (NDRS):** for any z and any $\alpha \geq 1$,

$$f(\alpha z) \geq \alpha f(z).$$

- **Constant returns to scale (CRS):** for any z and any $\alpha > 0$,

$$f(\alpha z) = \alpha f(z).$$

- **Euler equation (CRS, differentiable):**

$$f(z) = \sum_{i=1}^m \frac{\partial f(z)}{\partial z_i} z_i.$$

3. Example: $f(\ell) = \ell^\beta$

For $\beta > 0$ and $\alpha > 0$,

$$f(\alpha \ell) = (\alpha \ell)^\beta = \alpha^\beta \ell^\beta.$$

Hence:

- NIRS holds when $\beta \leq 1$.
- NDRS holds when $\beta \geq 1$.
- CRS holds when $\beta = 1$.

4. Profit maximization problem (PMP) and profit function

- **General PMP:** with prices $p = (p_1, \dots, p_n)$, a competitive producer solves

$$\max_{y \in Y} p \cdot y.$$

- **Single-product PMP:** with output price $p > 0$, input prices $w_1, \dots, w_m > 0$,

$$\max_{q \geq 0, z \geq 0} \left(pq - \sum_{i=1}^m w_i z_i \right) \quad \text{s.t.} \quad q \leq f(z).$$

- The maximized value is the **profit function** $\pi(\cdot)$; the set of maximizers is the **supply correspondence** $y(p)$.

5. Profit function properties and Hotelling's lemma

Let $\pi(p)$ be the profit function for the (multi-good) PMP.

- **Homogeneity of degree 1:** $\pi(\alpha p) = \alpha \pi(p)$ for $\alpha > 0$.
- **Convexity:** $\pi(\cdot)$ is convex (equivalently, $-\pi$ is concave).
- If π is differentiable and supply is single-valued, then **Hotelling's lemma** holds:

$$\frac{\partial \pi(p)}{\partial p_i} = y_i(p).$$

6. Implication of NDRS: profit is 0 or $+\infty$

If f exhibits nondecreasing returns to scale and there exists a plan with strictly positive profit, scaling it up yields arbitrarily large profit; otherwise the best attainable profit is 0.

7. Worked PMP example: $\max_{x, \ell} px - w\ell$ s.t. $x \leq \ell^\beta$, $x, \ell \geq 0$

- If $\beta \in (0, 1)$ (decreasing returns), the solution is interior and the notes derive

$$\ell^* = \left(\frac{\beta p}{w} \right)^{\frac{1}{1-\beta}}, \quad x^* = \left(\frac{\beta p}{w} \right)^{\frac{\beta}{1-\beta}},$$

with $\pi(p, w) = px^* - w\ell^*$.

- If $\beta = 1$ (CRS), then:
 - if $p < w$, optimal is $(x^*, \ell^*) = (0, 0)$ and $\pi = 0$;
 - if $p = w$, any (x, ℓ) with $x = \ell$ is optimal and $\pi = 0$;
 - if $p > w$, there is no finite maximizer and $\pi = +\infty$.
- If $\beta > 1$, the objective is not a concave program; even KKT points need not be global solutions (case skipped in the notes).

8. Cost minimization problem (CMP), cost function, and Shephard's lemma

Given target output $q \geq 0$ and input prices $w = (w_1, \dots, w_m) \gg 0$:

$$\min_{z \geq 0} \sum_{i=1}^m w_i z_i \quad \text{s.t.} \quad f(z) \geq q.$$

- Minimizers define the **conditional factor demand** correspondence $z(w, q)$.
- The minimized value is the **cost function** $C(w, q)$.
- Properties:
 - $C(\alpha w, q) = \alpha C(w, q)$ (degree 1 in w),
 - C is nondecreasing in q ,
 - $C(\cdot, q)$ is concave in w ,
 - $z(\alpha w, q) = z(w, q)$ (degree 0 in w).
- **Shephard's lemma:** if differentiable,

$$\frac{\partial C(w, q)}{\partial w_i} = z_i(w, q).$$

9. Linking CMP and PMP

If the cost function $C(w, q)$ is known, the single-output PMP can be written as:

$$\max_{q \geq 0} (pq - C(w, q)).$$

Session 2: Competitive Equilibrium, Pareto Efficiency, and Welfare Theorems

1. General equilibrium environment

- There are I consumers and J producers, with n goods. Markets are competitive (all agents take prices as given).

2. Consumer and producer problems

- **Consumer i :** chooses $x_i = (x_{i1}, \dots, x_{in})$ to solve

$$\max_{x_i \geq 0} u_i(x_i) \quad \text{s.t.} \quad p \cdot x_i \leq m_i.$$

Income has the form

$$m_i = p \cdot e_i + \sum_{j=1}^J \theta_{ij} \pi_j,$$

where e_i is endowment and $\theta_{ij} \in [0, 1]$ is consumer i 's ownership share of firm j .

- **Producer j :** chooses $y_j \in Y_j$ to solve

$$\max_{y_j \in Y_j} p \cdot y_j,$$

with optimal profit denoted $\pi_j(p)$.

3. Competitive equilibrium

A price vector $p^* \gg 0$ and allocation (x^*, y^*) form a competitive equilibrium if:

- (1) Given p^* , each consumer i chooses x_i^* solving their UMP.
- (2) Given p^* , each producer j chooses y_j^* solving their PMP, with profits $\pi_j(p^*)$.
- (3) **Market clearing:** for each good k ,

$$\sum_{i=1}^I x_{ik}^* = \sum_{i=1}^I e_{ik} + \sum_{j=1}^J y_{jk}^*.$$

4. Example: Robinson Crusoe economy (2 goods: consumption x and labor ℓ)

- One consumer with utility

$$u(x, \ell) = \log x + b \log(1 - \ell), \quad b > 0,$$

endowed with 1 unit of labor.

- One firm (owned by the consumer) with production

$$f(\ell) = \ell^\alpha, \quad \alpha \in (0, 1).$$

- Let p be the price of consumption and w the wage (price of labor). Equilibrium requires $p > 0$ and $w > 0$.

- Producer optimal choices (from the PMP solution):

$$\ell^{D*} = \left(\frac{\alpha p}{w}\right)^{\frac{1}{1-\alpha}}, \quad x^{S*} = (\ell^{D*})^\alpha = \left(\frac{\alpha p}{w}\right)^{\frac{\alpha}{1-\alpha}},$$

with positive profit π .

- Consumer chooses (x^D, ℓ^S) with $0 \leq \ell^S \leq 1$ and budget $px^D \leq w\ell^S + \pi$; solving yields closed-form expressions in (p, w, π, b) .
- Market clearing imposes $\ell^{S*} = \ell^{D*}$ and $x^{D*} = x^{S*}$, which pins down equilibrium prices up to scale (a condition on p/w).

5. Example: Edgeworth box economy (2 consumers, 2 goods, no producers)

- Utilities:

$$\begin{aligned} u_1(x_{11}, x_{12}) &= \alpha \log x_{11} + (1 - \alpha) \log x_{12}, & \alpha &\in (0, 1), \\ u_2(x_{21}, x_{22}) &= \beta \log x_{21} + (1 - \beta) \log x_{22}, & \beta &\in (0, 1). \end{aligned}$$

- Endowments: consumer 1 has $(10, 0)$; consumer 2 has $(0, 10)$.
- Solving demands and imposing market clearing yields the equilibrium relative price:

$$\frac{p_1}{p_2} = \frac{1 - \alpha}{\beta},$$

and an equilibrium allocation consistent with $x_{11}^* + x_{21}^* = 10$ and $x_{12}^* + x_{22}^* = 10$.

6. Pareto efficiency and social planner formulation

- A feasible allocation is **Pareto efficient** if no other feasible allocation makes every consumer weakly better off and at least one strictly better off.
- **Weighted social welfare problem:** choose allocations to maximize

$$\lambda_1 u_1(x_1) + \cdots + \lambda_I u_I(x_I), \quad \lambda_i \in [0, 1], \quad \sum_i \lambda_i = 1,$$

subject to feasibility. Varying the weights generates different Pareto efficient allocations.

7. Pareto set in the Edgeworth box example

Let $\lambda \in [0, 1]$ be the weight on consumer 1. The notes derive the Pareto efficient allocations:

$$\begin{aligned} x_{11}^* &= \frac{10 \lambda \alpha}{\lambda \alpha + (1 - \lambda) \beta}, & x_{12}^* &= \frac{10 \lambda (1 - \alpha)}{\lambda (1 - \alpha) + (1 - \lambda) (1 - \beta)}, \\ x_{21}^* &= 10 - x_{11}^*, & x_{22}^* &= 10 - x_{12}^*. \end{aligned}$$

Special case $\alpha = \beta = 0.5$: competitive equilibrium allocation is $(5, 5, 5, 5)$ and the Pareto set is

$$x_{11} = x_{12} = 10\lambda, \quad x_{21} = x_{22} = 10(1 - \lambda),$$

which coincides with the competitive equilibrium at $\lambda = 0.5$.

8. Welfare theorems (statements)

- **First welfare theorem:** a competitive equilibrium allocation is Pareto efficient.
- **Second welfare theorem:** any Pareto efficient allocation can be supported as a competitive equilibrium with appropriate redistribution of endowments (prices plus a feasible endowment distribution that decentralizes the desired allocation).